Factsheet: Type I and Type II Error

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Summary

An introduction to Type I and Type II error rates used in hypothesis testing.

When conducting hypothesis tests we first assign a α error rate (otherwise known as significance level). $1-\alpha$ is your confidence level, so how statistically certain are you of the outcome of your hypothesis test.

For example, at $\alpha = 0.05$ you have a 5% probability of rejecting your null hypothesis (H_0) when it is in fact true. This is otherwise known as the **Type I error**.

Conversely, if you're failing to reject the null hypothesis where it is false, this is called **Type II error**.

Commonly it is the **Type I error** that is first fixed and only then is the **Type II error** rate calculated.

> **Example 1**
>
> Cantor's Confectionery checks if their trademark biscuits are always of radius 4 centimeters, as advertised. To that end, he decides to use a t-test. Wanting to have high certainty, but at the same time to pin down where the true might lie (associated confidence intervals), he decides to fix the **Type I error** level at $\alpha = 0.05$ (5%), and thus making his confidence level equal to $1 - \alpha = 1 - 0.05 = 0.95$ (95%)

> Definition of Type I and II error.
>
> - **Type I error (false negative)**: the probability of rejecting the null hypothesis (H_0) when it is true.
> - **Type II error (false positive)**: the probability of failing to reject the null hypothesis (H_0) when it is false.

Summarized in a table:

	H_0 is true	H_0 is false
Reject H_0	Type I error (α)	Correct decision $(1-\beta)$

	H_0 is true	H_0 is false
Fail to reject H_0	Correct decision $(1-\alpha)$	Type II error (β)

Tip

Statistical power is a measure of the ability to detect differences (between parameters) / effects (in the case of regression), given that they actually exist.
In other words, it is the measure of correctly failing to reject the null hypothesis (H_0) when it is in fact true, so in mathematical notation - your statistical power is: $1-\beta$.
You can also think of it as the probability of **NOT** making a Type II error.
Conventionally, statistical power equal to 0.8 is thought to be an acceptable power threshold. This is because of practical reasons: increasing statistical power requires substantial expansion of sample sizes, while the resulting reduced Type II error rate is usually thought of as not compensating the costs associated with increasing sampling.

Calculating Type II error requires more work and you should not attempt to do it by hand. Therefore, below you can find a Type II error calculator for four most common hypothesis tests (Z-test, t-test, ANOVA, and χ^2 test):

```
#| '!! shinylive warning !!': |
#|   shinylive does not work in self-contained HTML documents.
#|   Please set `embed-resources: false` in your metadata.
#| standalone: true
#| viewerHeight: 800
library(shiny)
library(bslib)

ui <- page_sidebar(
  title = "Type II Error Calculator",
  sidebar = sidebar(
    selectInput("test_type", "Select Test:",
                choices = c("Z Test" = "z",
                           "t Test" = "t",
                           "ANOVA" = "anova",
                           "Chi-Squared" = "chisq")),

    numericInput("alpha", "Significance Level ( ):", 0.05, min = 0.001, max = 0.5),
```

```
    conditionalPanel(
      condition = "input.test_type == 'z' || input.test_type == 't'",
      numericInput("n", "Sample Size:", 30, min = 2),
      numericInput("mu0", "Null Mean ( ):", 0),
      numericInput("mu1", "True Mean ( ):", 0.5),
      numericInput("sigma", "Std Dev ( ):", 1, min = 0.01)
    ),

    conditionalPanel(
      condition = "input.test_type == 'anova'",
      numericInput("k", "Number of Groups:", 3, min = 2),
      numericInput("n_per_group", "Sample per Group:", 20, min = 2),
      textInput("group_means", "Group Means:", "0, 0.3, 0.6"),
      numericInput("sigma_anova", "Std Dev ( ):", 1, min = 0.01)
    ),

    conditionalPanel(
      condition = "input.test_type == 'chisq'",
      numericInput("total_n", "Sample Size:", 100, min = 10),
      numericInput("df", "Degrees of Freedom:", 2, min = 1),
      textInput("null_probs", "Null Probs:", "0.33, 0.33, 0.34"),
      textInput("true_probs", "True Probs:", "0.25, 0.35, 0.40")
    )
  ),

  card(
    card_header("Decision Table"),
    uiOutput("decision_table")
  ),

  card(
    card_header("Results"),
    verbatimTextOutput("results")
  )
)

server <- function(input, output) {
```

```r
calc <- reactive({
  res <- list(beta = NA, power = NA, error = NULL)

  if (input$test_type == "z") {
    se <- input$sigma / sqrt(input$n)
    z_crit <- qnorm(1 - input$alpha/2)
    crit_upper <- input$mu0 + z_crit * se
    crit_lower <- input$mu0 - z_crit * se

    res$beta <- pnorm(crit_upper, input$mu1, se) - pnorm(crit_lower, input$mu1, se)
    res$power <- 1 - res$beta

  } else if (input$test_type == "t") {
    pw <- power.t.test(n = input$n,
                       delta = input$mu1 - input$mu0,
                       sd = input$sigma,
                       sig.level = input$alpha,
                       type = "one.sample")
    res$power <- pw$power
    res$beta <- 1 - res$power

  } else if (input$test_type == "anova") {
    means <- as.numeric(strsplit(input$group_means, ",")[[1]])

    if (length(means) != input$k) {
      res$error <- "Group means count must match k"
      return(res)
    }

    grand_mean <- mean(means)
    f <- sqrt(sum((means - grand_mean)^2) / input$k) / input$sigma_anova

    df1 <- input$k - 1
    df2 <- input$k * (input$n_per_group - 1)
    lambda <- input$n_per_group * input$k * f^2

    f_crit <- qf(1 - input$alpha, df1, df2)
    res$power <- 1 - pf(f_crit, df1, df2, ncp = lambda)
```

```
    res$beta <- 1 - res$power

  } else if (input$test_type == "chisq") {
    null_p <- as.numeric(strsplit(input$null_probs, ",")[[1]])
    true_p <- as.numeric(strsplit(input$true_probs, ",")[[1]])

    if (abs(sum(null_p) - 1) > 0.01 || abs(sum(true_p) - 1) > 0.01) {
      res$error <- "Probabilities must sum to 1"
      return(res)
    }

    w <- sqrt(sum((true_p - null_p)^2 / null_p))
    lambda <- input$total_n * w^2

    chi_crit <- qchisq(1 - input$alpha, input$df)
    res$power <- 1 - pchisq(chi_crit, input$df, ncp = lambda)
    res$beta <- 1 - res$power
  }

  res
})

output$decision_table <- renderUI({
  res <- calc()

  if (!is.null(res$error)) {
    return(p(strong("ERROR: "), res$error))
  }

  tagList(
    tags$table(
      class = "table table-bordered",
      style = "width: 100%; text-align: center;",
      tags$thead(
        tags$tr(
          tags$th(""),
          tags$th(HTML("H<sub>0</sub> is true")),
          tags$th(HTML("H<sub>0</sub> is false"))
```

```
        )
      ),
      tags$tbody(
        tags$tr(
          tags$td(HTML("Reject H<sub>0</sub>")),
          tags$td(HTML(paste0("Type I error ( )<br>", round(input$alpha, 4)))),
          tags$td(HTML(paste0("Correct decision (1 -  )<br>", round(res$power, 4))))
        ),
        tags$tr(
          tags$td(HTML("Fail to reject H<sub>0</sub>")),
          tags$td(HTML(paste0("Correct decision (1 -  )<br>", round(1 - input$alpha,
          tags$td(HTML(paste0("Type II error ( )<br>", round(res$beta, 4))))
        )
      )
    )
  )
})

output$results <- renderPrint({
  res <- calc()

  if (!is.null(res$error)) {
    cat("ERROR:", res$error)
    return()
  }

  cat("Type II Error ( ):", round(res$beta, 4), "\n")
  cat("Power (1 -  ):", round(res$power, 4), "\n\n")
  cat("Interpretation:\n")
  cat(round(res$beta * 100, 1), "% chance of missing a real effect\n")
  cat(round(res$power * 100, 1), "% chance of detecting a real effect")
})
}

shinyApp(ui, server)
```

Example 2

Cantor's Confectionery is conducting a t-test to check if their famous chocolate bars are indeed always 75 grams in weight, as Mr. Cantor suspects that his machine might be malfunctioning.
The null and alternative hypotheses are:

- $H_0 = 75$
- $H_1 \neq 75$

Mr. Cantor employs a statistician who decides to test at 95% confidence level, therefore allowing for a Type I error rate of
Additionally, he learns from Mr. Cantor that the sample sizes of the chocolate bars will be 25 and there will be two batches to compare. Therefore the statistician decides to adopt a t-test.
After obtaining the raw data, the statistician calculates the mean and the standard deviation:

$$\bar{x} = 75.019$$

$$\sigma = 1.0259$$

Now, he can use the Type II error calculator to complete the table of Type I and Type II errors:

	H_0 is true	H_0 is false
Reject H_0	Type I error ($\alpha = 0.05$)	Correct decision ($1 - \beta = 1 - 0.0307 = 0.9693$)
Fail to reject H_0	Correct decision ($1 - 0.05 = 0.95$)	Type II error ($\beta = 0.0307$)

Further reading

For more information on hypothesis test, please go to Guide: Hypothesis testing For more information on t-test, please go to Guide: T-testing For information on ANOVA, please go to Guide: ANOVA

Version history

v1.0: initial version created 04/26 by Jakub Brzozowski as a part of a University of St Andrews VIP project.

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